

Search formulation

Systematic search to solve a problem through partial assignment -
Assignment to all the variables in the search space indicates to local search

Initially no assignments

CSP (Constraint Satisfaction Problem) -

Set of variables $X = \{x_1, x_2, \dots, x_n\}$

Each variable has its own domain

Set of domain $D = \{D_1, D_2, \dots, D_n\}$

Set of constraints $C = \{c_1, c_2, \dots, c_k\}$

S - scope of constraint
 $S_i \subseteq X$

$$C_i = (S_i, R_i)$$

Constraints - specifies allowable combination of variables
 $S_i = \{x_{i1}, x_{i2}, \dots, x_{ip}\}$

Relation $R_i \subseteq (D_{i1} \times D_{i2} \times \dots \times D_{ip})$

4 queen problem is a CSP problem [constraints - one queen should not attack other]

	x_1	x_2	x_3	x_4
1				
2				
3				
4				

Constraints between x_1 and x_2

$$R_{12} = \{(1,3), (1,4), (2,4), (3,1), (4,1), (4,2)\}$$

These are the pair of values that the variables can take

Similarly $R_{13}, R_{14}, R_{24}, R_{34}$ can be the other constraints

Scope - Set of variables that participate in constraints

An assignment $\bar{a}$ over scope S

$$x_1 = a, x_2 = b$$

Assignment selects each variable from its respective domain

Relation - defines values that a variable can take

Scope means giving values against variables

Satisfies a constraint C_i at $S_i \subseteq S$

$$\text{Projection } \pi_{S_i} \bar{a} \subseteq R_i$$

This $\bar{a}$ satisfy relation R_{12} (as 1,4 common)

	x_1	x_2	x_3	x_4
1	Q			
2			Q	
3				
4		Q		

$$\bar{a} = \{1, 4, 2\}$$

CSP is solved using Intelligent Backtracking algorithm.

8 queen problem

Let, consider 4 queen problem on chess board having dimension of 4×4

The constraint ~~are~~ is no queen should be under attack.

Thus row, column, diagonally should not be the same

Hence possible solution is $16C_4$

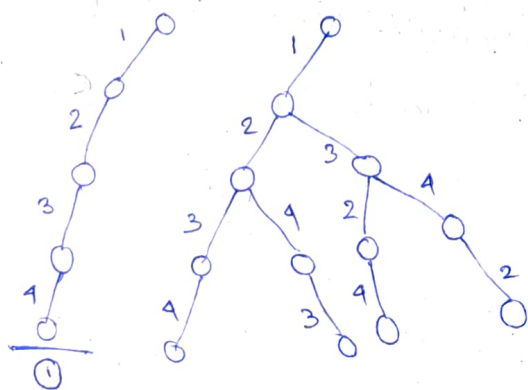
Then to reduce the problem it is considered that one queen will be placed in one column, also not in same row.

Hence this type problem is solved through backtracking algorithm.

In this context state space tree is used. The state space tree represents all the possible solution in the problem space.

Backtracking follows depth first search

Ans
↓
Queens



Hence by moving the possible positions of the queens, the solution is reached

	1	2	3	4
1	Q ₁			
2		Q ₂		
3			Q ₃	
4				Q ₄

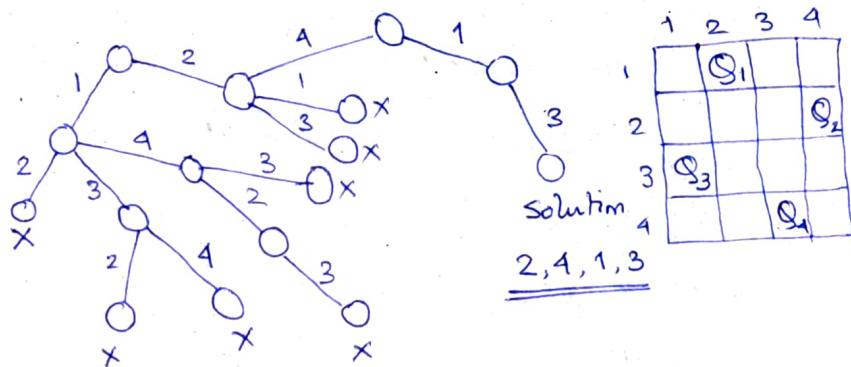
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In this case, Only row/column conditions are considered but diagonal attack has not been taken into account.

Hence generation of total nodes

$$= 1 + \sum_{i=1}^4 \left[\prod_{j=0}^i (N-j) \right]$$

Hence applying Bounding Condition $\Rightarrow$ Same Row/Column/Diagonal



So, in the above search process, any time the bounding condition is not matched, the node is killed in the tree (further not expanded). Then the search backtracks ~~and~~ to previous node and starts expanding with other options. Hence through this searching the solution is reached.